

tion, one should focus on augmenting the skills required for embedded system development. For example, a person with electronics background should focus more on programming, software engineering and real-time system skills," explains Chandrakant Sakharwade, general manager, embedded division, DKOP Labs, from his vast experience as a trainer in this field.

Tips from the experts...

Your next question is likely to be "How to prepare myself for this niche field?" Let us ask the experts in this field.

Ajit Clarence, head, EFY Tech Center, Bengaluru, says, "As an embedded engineer, it is very important to have knowledge of hardware and software, which sets one apart from a normal software programmer. This is one reason why embedded engineers are paid more than normal software engineers."

From his varied experience in embedded industry and training, Clarence feels that the curriculum taught in various universities is more of theoretical in nature. Therefore he makes his students industry-ready by giving importance on hands-on training, both in hardware and software.

Clarence says that, an embedded engineer should have interest in basic electronics. He should have rigged up at least one circuit and tested it. He should be able to identify electronic components available in the market and know how to use them in his designs.

In fact, the challenge lies in understanding the electronics and other interface hardware. Understanding the fundamentals of embedded systems and knowledge of verification tools in an IDE setting are required.

Given that most electronic devices use microprocessors or microcontrollers, try to get in-depth knowledge of microcontrollers and printed circuit boards. "Programming, debugging and testing of microcontrollers as well as knowledge of printed circuit boards with respect to schematics and routing

are must-to-have skills," informs Clarence.

Clarence also points out that electronic circuits (analogue designs) are very tricky, and you should be very observant and patient to avoid damage to the system that you design.

Kumar says that, the industry expects you to have super specialisations within the field, such as Embedded C programming, exploring controller-specific features under Embedded C, RTOS porting and application development, knowledge of special networking protocols, along with sound understanding of fundamentals like eight-bit microprocessors/microcontrollers, electronics hardware (basic, digital, analogue) and PCB designing.

Patel emphasises on the need of a broad approach to education as embedded systems encompass a wide range of applications, technologies and disciplines. He lays stress on the importance of 'C' language (as most of embedded work is based on it), small and single-microcontroller applications, control systems, distributed embedded control, system-on-chips, networking, embedded PCs, critical systems, robotics, computer peripherals, wireless data systems, signal processing, and command and control. Additional cross-enterprise skills that are important for embedded system designers include security, dependability, energy-aware computing, system engineering, real-time computing and human-computer interaction.

One should be able to think logically and learn to use real-time operating systems. Some of the advanced topics like VLSI, DSP and robotics need more knowledge of topics like analogue and digital designs, signal analysis and mechanical design, respectively.

The last frontier

It is possible that your awareness about most of the aforementioned terms is from a notional perspective only. The industry, however, needs something extra—the actual in-hand experience. Various private- and public-sector or-

ganisations absorb final-year and pre-final-year students in their live projects.

Patel informs, "At elfnochips, we recruit students at fresher level through campus placement wherein they start their eighth-semester industrial training. On an average, hiring happens at the entry level, in ASIC, embedded and software development divisions. As mentioned before, we provide technical and non-technical trainings to make freshers competent enough to work at elfnochips on customer projects either by owning a module or work with senior engineers."

Similarly, at KPIT Cummins, trainees spend initial 3-6 months on learning the core essentials and then move to roles of software engineer in the areas of electronic design and software development.

On the other hand, industry-academia partnership models like TI's 'University Relationship' and Infochips' 'Impetus' are good endeavours to make students industry-ready.

Ganesh Guruswamy, vice president and country manager, Freescale Semiconductor, opines, "While embedded software companies are leveraging the growing availability of fresh engineering graduates, they must also spend considerable time and resources on training and boosting productivity. Currently, our universities offer limited exposure to real-world situations, and this poses a challenge when we hire fresh graduates. Increased industry participation in talent development will ensure industry-ready talent."

Become a go-getter

So dig out all possibilities to get a practical exposure. At the same time, keep abreast of the latest technological trends through newspapers, technical magazines, technical seminars and, last but not the least, trade fairs. You can learn about 'chip to ship' of an embedded system only after completing a proper project. ●

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